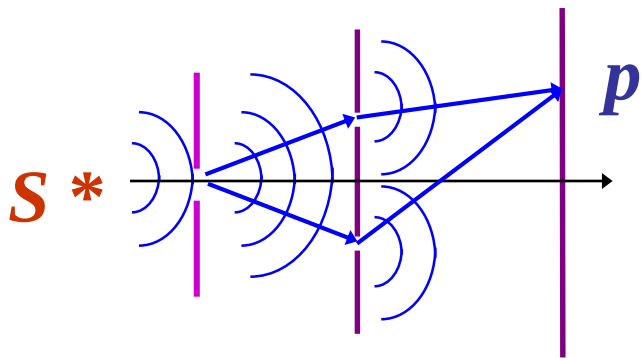
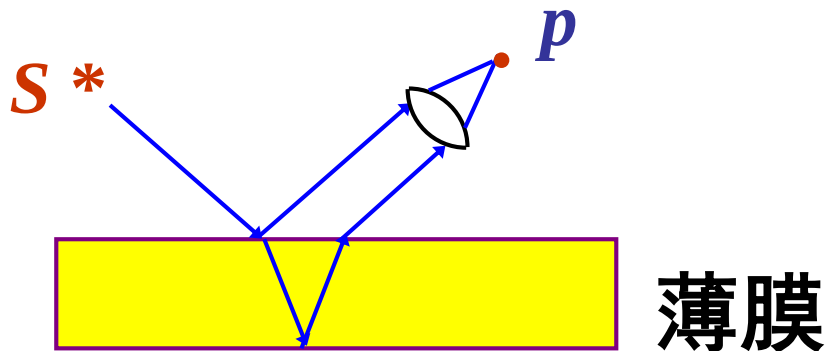


3.2 分波阵面干涉

普通光源获得相干光的途径



分波面法



分振幅法

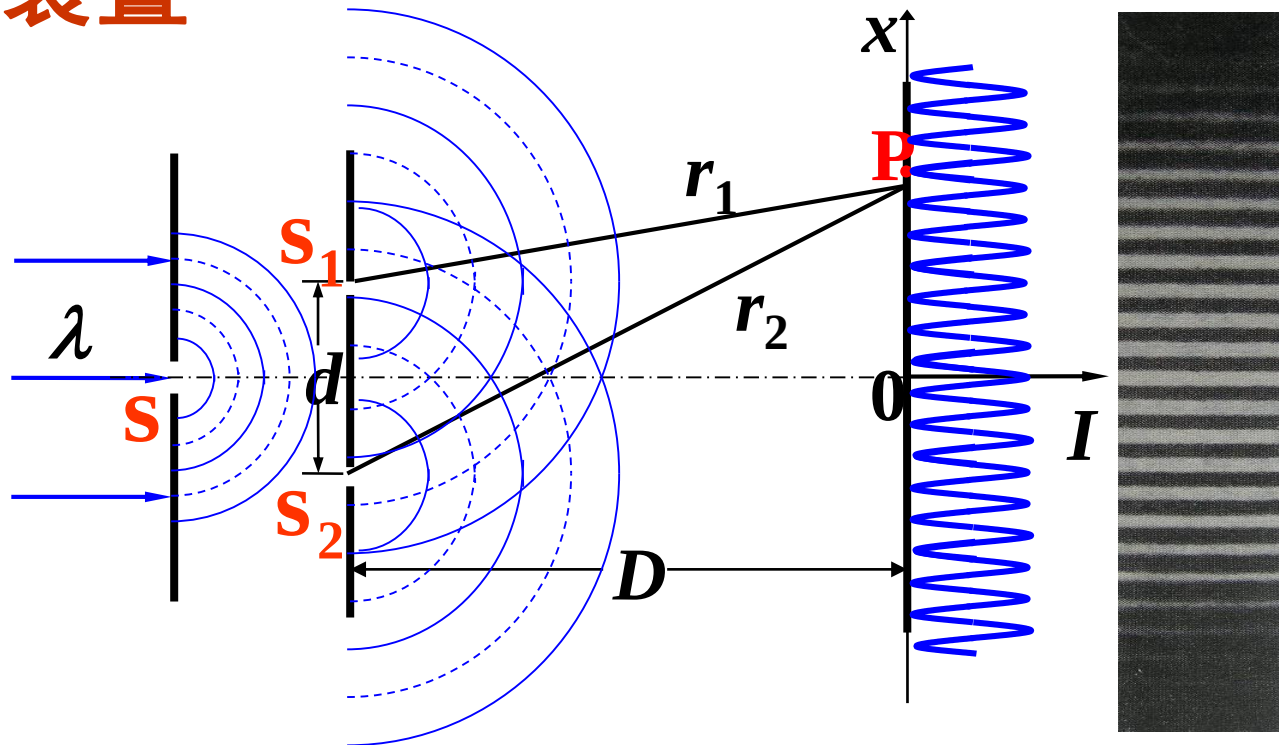
3.2.1.杨氏双缝干涉

英国物理学家托马斯·杨（1773-1829）使用巧妙的技术从单个光源获得两个相干光源，首次实现了可见光干涉。证明了光的波动性，还首次测量了光的波长



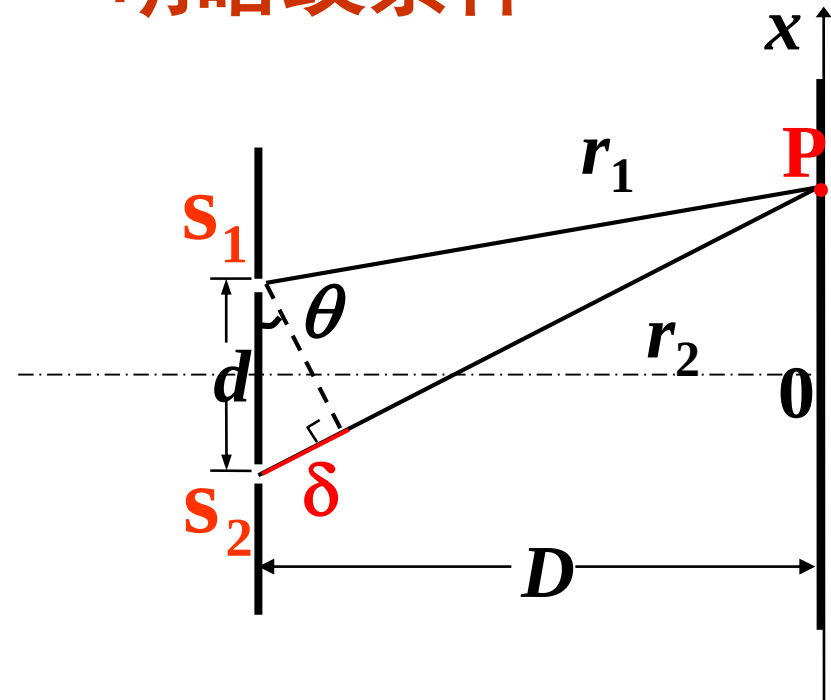
托马斯·杨 (Thomas Young)
1773-1829, 英国物理学家

实验装置



图中：相邻实线与，虚线的相位差为 π

明暗纹条件



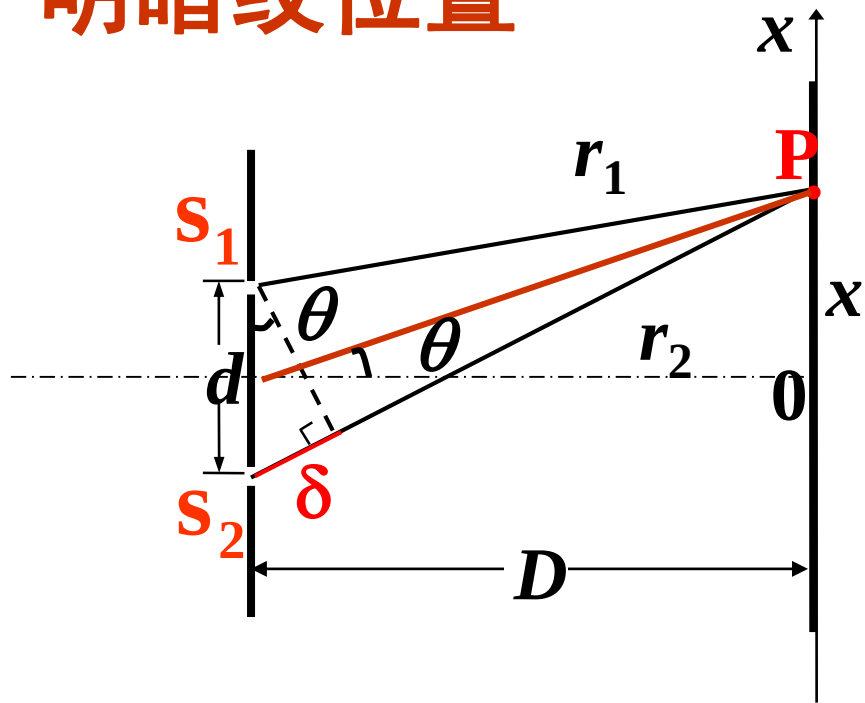
$$\delta = r_2 - r_1 = \pm k\lambda$$

$$\delta = r_2 - r_1 = \pm(2k-1)\frac{\lambda}{2}$$

$$d \sin \theta = \pm k\lambda$$

$$d \sin \theta = \pm(2k-1)\frac{\lambda}{2}$$

明暗纹位置



$$d \sin \theta = \pm k \lambda$$

$$d \sin \theta = \pm (2k - 1) \frac{\lambda}{2}$$

$$x = D \tan \theta \approx D \sin \theta$$

明纹中心

$$x = D \operatorname{tg} \theta \approx D \sin \theta \quad d \sin \theta = \pm k \lambda$$

明纹中心

$$x = D \sin \theta = D \left(\pm \frac{k}{d} \right) \lambda, k = 0, 1, 2, \dots$$

$$x = \pm k \frac{D}{d} \lambda$$

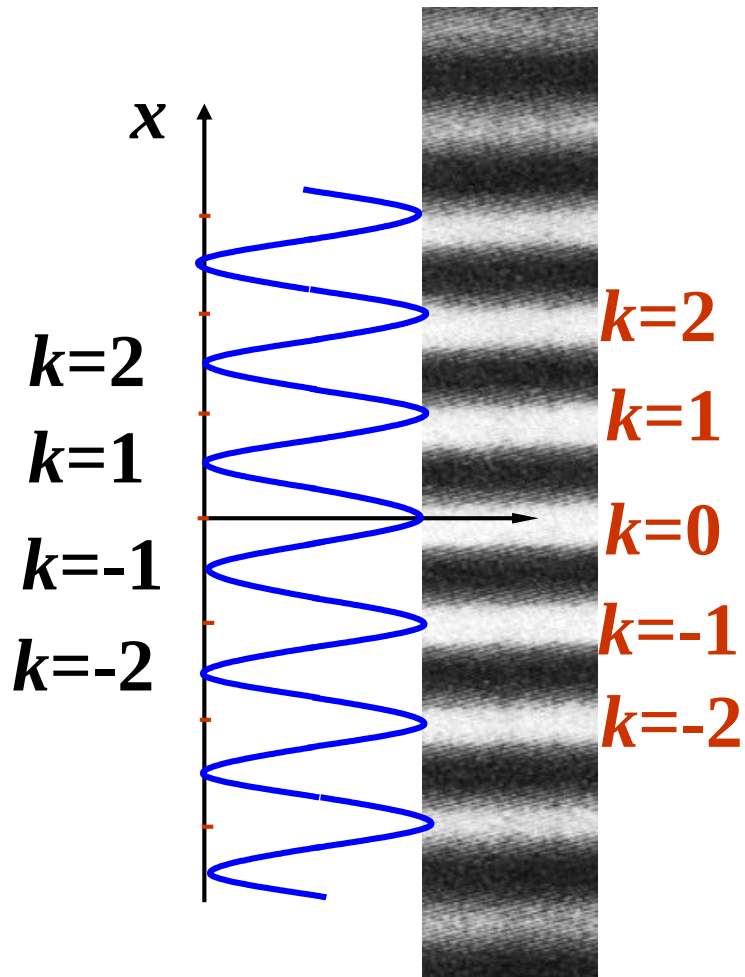
暗纹中心

$$x = D \sin \theta \quad d \sin \theta = \pm (2k - 1) \frac{\lambda}{2}$$

$$= D \left[\pm (2k - 1) \frac{\lambda}{2d} \right]$$

$$k = 1, 2, 3, \dots$$

$$x = \pm (2k - 1) \frac{D}{2d} \lambda$$



明纹中心

$$x = \pm k \frac{D}{d} \lambda$$

$$k=0,1,2,3\dots$$

暗纹中心

$$x = \pm (2k - 1) \frac{D}{2d} \lambda$$

$$k=1,2,3\dots$$



明纹中心 $x = \pm k \frac{D}{d} \lambda \quad k=0,1,2,3\dots$

暗纹中心 $x = \pm(2k-1) \frac{D}{2d} \lambda \quad k=1,2,3\dots$

相邻明（暗）纹间的距离

$$\Delta x = (k+1) \frac{D}{d} \lambda - k \frac{D}{d} \lambda \quad \Delta x = \frac{D}{d} \lambda$$

讨论:

$$\Delta x = \frac{D}{d} \lambda$$

- 1) θ 不太大时, Δx 与 k 无关, 条纹等间距排列
- 2) 波长 λ 一定, $\Delta x \propto D, \Delta x \propto 1/d$
- 3) D, d 一定, $\Delta x \propto \lambda$

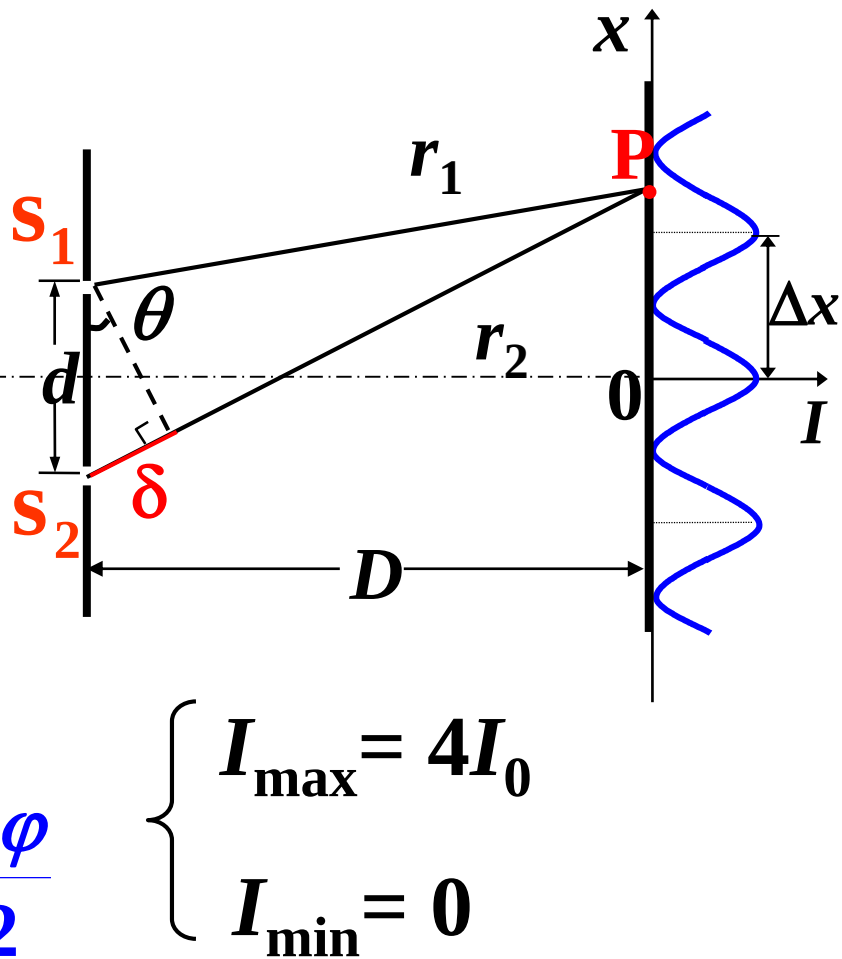
光强分布

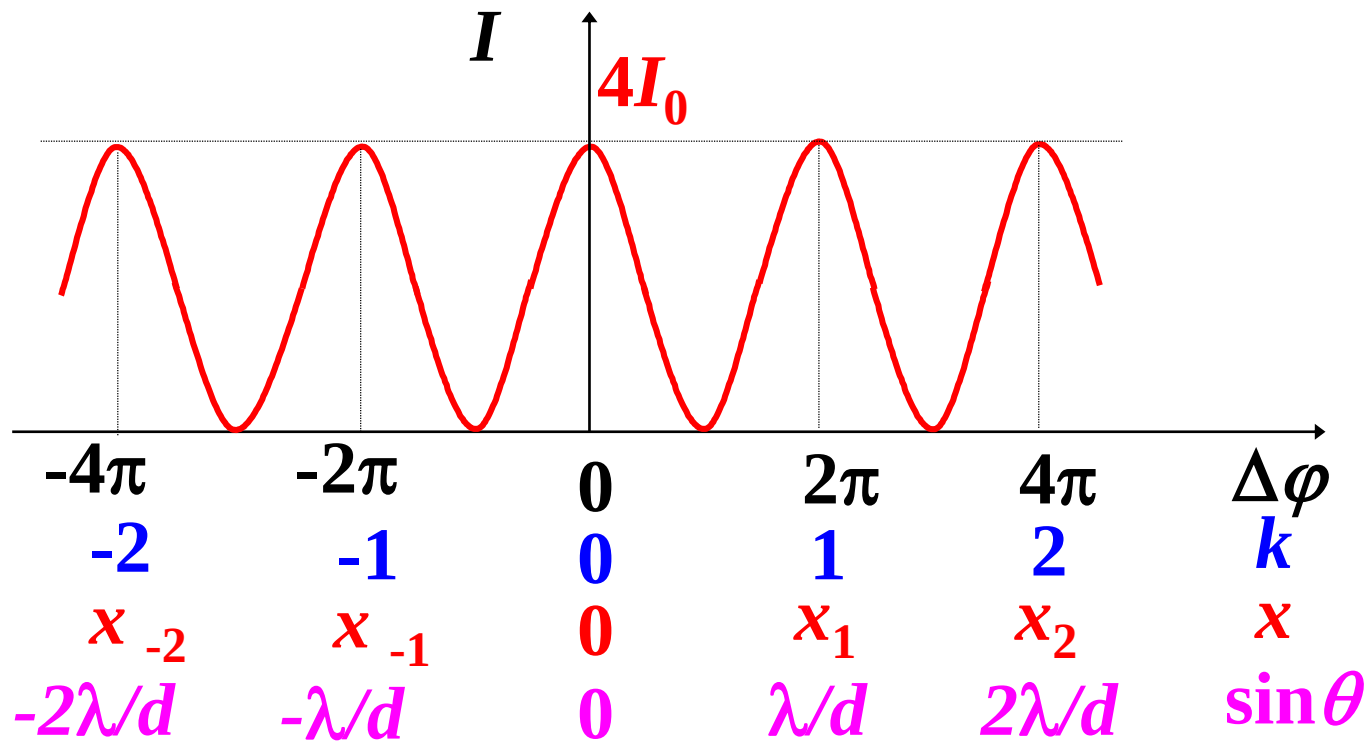
$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta\varphi$$

$$(\Delta\varphi = \frac{2\pi}{\lambda} d \sin\theta)$$

$$\begin{cases} I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} \\ I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2} \end{cases}$$

若 $I_1 = I_2 = I_0$ $I = 4I_0 \cos^2 \frac{\Delta\varphi}{2}$





入射光为复色光

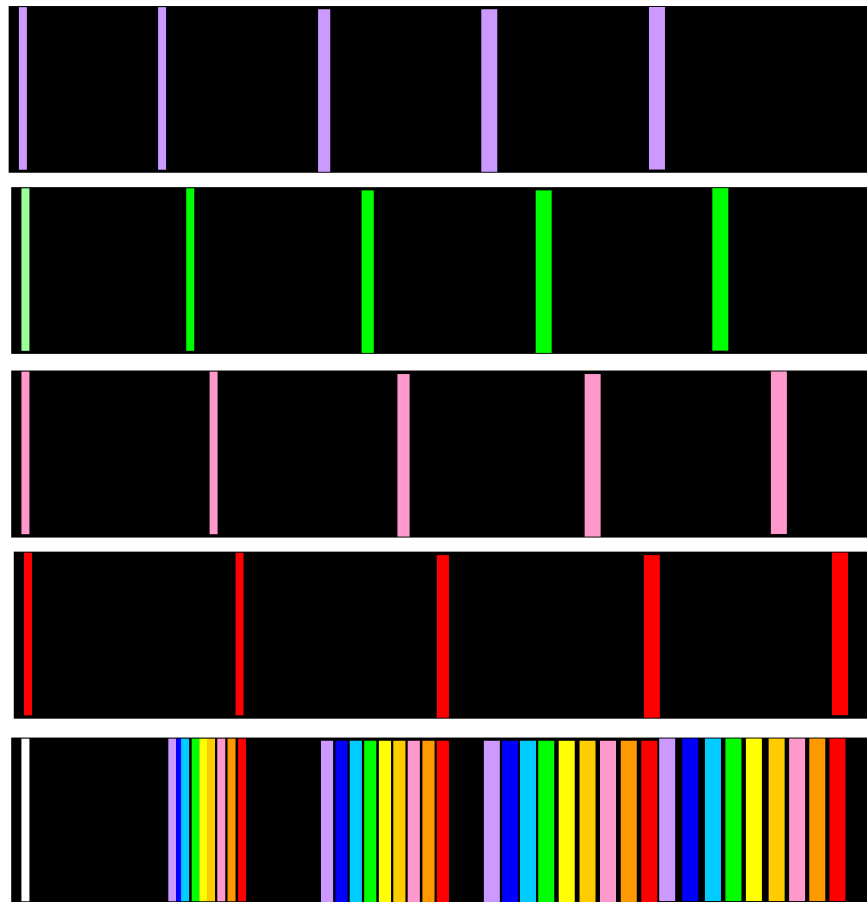
1) 各级明纹 (不含零级) 都是彩色条纹

$$x = k \frac{D}{d} \lambda$$

2) 中央零级明纹是白色

3) 高级次明纹可出现重合

$$\Delta x = \frac{D}{d} \lambda \quad \lambda \text{ 大 } \Delta x \text{ 大}$$



用白光作光源, λ 从7000 Å(红)到4000 Å(紫),观察双缝干涉条纹,设缝间距为 d ,清晰可见的光谱的级次.

$$\Delta x = \frac{D}{d} \lambda$$

红光的条纹间距大于紫光的条纹间距

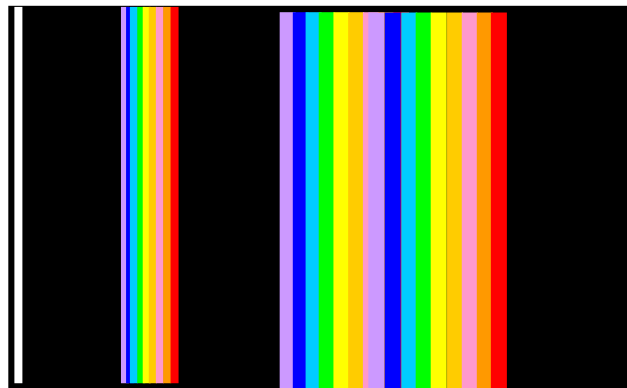
$$x = \pm k \frac{D}{d} \lambda \qquad x = \pm (k + 1) \frac{D}{d} \lambda$$

如果红光 $\lambda=7000\text{\AA}$ 第 k 级明纹正好是
第 $k+1$ 级紫光 $\lambda=4000\text{\AA}$ 的明纹

$$x_{\text{红}}(k) = x_{\text{紫}}(k + 1)$$

$$k\lambda_{\text{红}} = (k + 1)\lambda_{\text{紫}}$$

$$k = \lambda_{\text{紫}} / \Delta\lambda = 1.3$$



清晰可见的光谱只有0级和±1级